GenAI for Engineers Lesson Plan

IES | v3.1 | LP-aligned one-day delivery

Follow the supplied Lesson Plan with Hands-On Labs (v2 - PUB Feedback) module order and clock times. Retain its six modules; consolidate learner work into only 65 minutes of guided Excel exercises within Module 3. Other practical examples are trainer-led demonstrations or discussion. The original agent-build outcomes are demonstrated, not assessed as independent learner deployment.

09:00–17:00 = 480 minutes: 390 teaching minutes + 60 lunch + 30 tea. The source LP calls this seven training hours excluding lunch; that includes tea. This delivery plan uses 6.5 net teaching hours. Guided Excel time is 25 + 20 + 20 = 65 minutes, included within Module 3, never added on top.

Time	LP module	Delivery allocation
09:00–10:00	1 · Copilot ecosystem	15 overview; 10 access check; 20 Excel demonstration; 15 discussion
10:00–11:15	2 · Prompting for engineering documents	20 RISE/E-MAP; 20 document demonstration; 20 constraints and persona review; 15 prompt critique
11:15–11:30	Tea break	15 minutes
11:30–13:00	3 · Engineering methodology and Excel	15 primer; 25 guided SPC; 20 guided FMEA; 20 guided DMAIC; 10 shared review
13:00–14:00	Lunch	60 minutes
14:00–15:15	4 · Copilot and reasoning agents	15 concepts; 25 FMEA/report assistant demo; 20 research/analysis comparison; 15 questions and buffer
15:15–15:30	Tea break	15 minutes
15:30–16:15	5 · SharePoint grounding	10 grounding; 20 retrieval/citation demo; 15 permissions and discussion
16:15–16:45	6 · AI security and toolkit	20 seven-step review; 10 personal prompt toolkit
16:45–17:00	Closing and Q&A	10 recap; 5 next action

Only three short guided exercises

Use one working copy of Worked-Reference.xlsx throughout. Prepared formulas, charts and tables keep the exercise feasible for beginners. Starter.xlsx and the full prompt bank are for optional practice after the course. There is no requirement to finish them during class.

1 · Copilot ecosystem

Show the same engineering question in Chat and Excel. Identify which evidence is available in each surface. Complete the access check once; use a paired screen or the supplied worked reference when access cannot be resolved within ten minutes.

2 · Prompting for engineering documents

Teach RISE: Role, Input, Steps, Expectations; E-MAP: Engineering context, Metrics, Assumptions, Parameters. Demonstrate converting one case note into a short specification and report. Compare a vague prompt with an evidence-constrained prompt. Demonstrate reliability, operations and safety

perspectives, one challenge per role, and a moderator summary preserving disagreement. Participants critique the displayed prompt; no separate document pack is required.

4 · Copilot and reasoning agents

Trainer demonstrates a prepared FMEA/report assistant: bounded instructions, rating-scale knowledge, one test and one correction. Compare a cited research response with an analysis response using the case evidence. Discuss the minutes-to-actions-to-report pattern. No learner creates, deploys or shares an agent. If a tenant feature is unavailable, walk through the prepared instructions and worked example; identify this as a demonstration fallback.

5 · SharePoint grounding

Trainer shows a prepared approved library, a question, retrieved passage and cited answer. Open the citation and check revision, scope and permissions. Discuss how a grounded answer could support an engineering report. No learner creates a site, library or agent. If live access is unavailable, use a labelled document-retrieval walkthrough rather than claiming a live agent run.

6 · AI security and toolkit

Apply seven checks to one displayed workbook claim: classification; governance/access; secure prompting; output verification; human sign-off; traceability/audit; awareness and incident reporting. Treat instructions embedded in retrieved documents as data, not authority. Learners retain one reusable prompt and one review rule; no additional lab submission.

Module 3 · SPC | 25 minutes

5 min · Read the prepared analysis

Open Worked-Reference.xlsx and save a working copy. The trainer has prepared data cleaning, unit conversion, baseline formulas and chart. Locate RawTests, Analysis and SPC_Summary; keep the source and baseline limits unchanged.

12 min · Prompt and inspect

Use SPC3 from Prompts to explain the existing chart and identify outside-limit points. If editing is available, add a signal flag in a copy. Check TEST-073 and one baseline row against fixed limits 60.03757–63.96743 L/h. Do not rebuild the chart or calculate Cp/Cpk in class.

8 min · Verify and retain

Confirm 17 valid monitoring points outside the limits. Save one three-sentence interpretation: signal, evidence and limitation. An association does not establish the cause. Stop at the time limit.

Core prompt · Plot and interpret

Review the existing Analysis and SPC_Summary chart. Keep the prepared formulas and fixed baseline limits unchanged. Identify valid monitoring points outside the limits and check one example against the source row. Return only three sentences: observed signal, supporting evidence, and an unresolved limitation. Do not rebuild the chart or calculate capability.

Module 3 · FMEA | 20 minutes

4 min · Read the prepared FMEA

Continue with the same workbook. Open FMEA_Review and ScoringNote. The six rows and provisional scores are supplied; do not generate a new system or rescore every row.

10 min · Prompt and check priorities

Use FMEA2 to explain or sort a copy by RPN and flag S>=9. Recompute FM01: 7×5×4=140. Check FM04: S=9 and RPN=90 still require severity review. Keep IDs attached to their source rows.

6 min · Review one row

Write one evidence gap for FM01 and one action proposal. Keep residual ratings unassessed. The multi-persona debate is a trainer demonstration in Module 2, not an extra participant task.

Core prompt · Calculate and prioritise

Review the prepared FMEA_Review. Check the RPN formulas for FM01 and FM04, preserving mode IDs and provisional scores. Explain why the severity-9 row still needs review. Return one evidence gap and one proposed action for FM01; keep residual ratings unassessed. Do not rebuild the table or chart.

Module 3 · DMAIC | 20 minutes

4 min · Read the prepared rates

Open Rate_Comparison, WeeklyEvents, Pareto and DMAIC in the same workbook. This eight-week event log is separate from the 120-test investigation.

10 min · Prompt a short summary

Use DMAIC1 to review the prepared DMAIC table and rates. Independently check 93/1008=9.22619% and 11/336=3.27381%. Use the supplied Pareto chart; do not build a new chart or a full Word report.

6 min · State a decision boundary

Save three cells: measured change, limitation and next evidence. The absolute change is 5.95238 percentage points. Two concurrent changes and a short pilot do not prove a cause or sustained improvement.

Core prompt · Define and measure

Review the prepared Rate_Comparison, WeeklyEvents and DMAIC tables. Check baseline and pilot rates using summed deviations divided by summed opportunities. Return three cells: measured change, limitation and next evidence. Do not create a full report or a new chart, and do not claim a causal effect from the short pilot.

Outcome coverage

LO1 methodology: primer and guided Excel review. LO2 documentation: trainer examples and Excel outputs; no independent Word/PPT pack. LO3 prompting: framework, critique and three selected prompts. LO4–LO6 agents, reasoning and SharePoint: demonstration-level coverage only. LO7 security: review of the displayed case. Independent agent deployment is not a completion claim for this shortened delivery.

Presentation teaching sequence — v3.1

Module 1 introduces AI, Copilot and the pump case before workbook names. Module 2 moves from a vague prompt to RISE, E-MAP, constraints and a worked persona challenge. Module 3 explains each method before its guided exercise. Modules 4–5 explain agents and grounding before demonstrations. Module 6 applies security checks and closes. The original clock times and 65-minute exercise total are unchanged.

Using the expanded 185-slide deck

Slides 1–64 are the one-day teaching route. Slide 65 introduces optional reference material. Slides 66–185 contain 30 worked examples; they are not additional scheduled labs. Keep the original six-module timetable and the 25/20/20-minute guided Excel exercises.

Reference index: SPC: slides 66–97; FMEA: slides 98–121; DMAIC: slides 122–145; Prompting: slides 146–165; Agents: slides 166–173; Grounding: slides 174–181; Security: slides 182–185

Each reference example follows data, prompt, illustrative answer and verification. Supporting CSVs, prompts and Worked-Examples.pdf are in Reference-Examples. The trainer demonstrations use the supplied Trainer-Demonstrations pack.